

CSC3066

Deep Learning

Analyzing Performance of ANNs

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What's to come today

- ❑ Plotting model learning curves
- ❑ Diagnosing model behaviour

Neural Network:

Plotting model learning curves

Diagnosing Model Behavior

- ❑ Plotting Model Learning Curves
 - ❑ Learning curve is a plot of Neural Network model performance over time calculated at the end of each training epoch.
 - ❑ Plots of learning curves provide insight into the learning dynamics of the model, such as whether the model is learning well, whether it is underfitting the training dataset, or whether it is overfitting the training dataset.

Diagnosing Model Behavior

- Plotting Model Learning Curves

- **Train Learning Curve:** Learning curve calculated from the training dataset that gives an idea of how well the model is learning.
- **Validation Learning Curve:** Learning curve calculated from a hold-out validation dataset that gives an idea of how well the model is generalizing.

Diagnosing Model Behavior

- Plotting Model Learning Curves
 - **Optimization Learning Curves:** Learning curves calculated on the metric by which the parameters of the model are being optimized, e.g. loss.
 - **Performance Learning Curves:** Learning curves calculated on the metric by which the model will be evaluated and selected, e.g. accuracy.

Diagnosing Model Behavior

- ❑ Plotting Model Learning Curves: [Validation Learning Curve](#)
 - ❑ Reference to a validation dataset can be included in the call to the `fit` function.
 - ❑ Validation set is a portion of the training set not used to fit the model, and is instead used to evaluate the performance of the model during training.
 - ❑ The data can be split manually and specified by the `validation_data` argument, or the split can be done automatically via the `validation_split` argument.

Diagnosing Model Behavior

- Plotting Model Learning Curves: Validation Learning Curve
 - The `fit` function will return a `History` object that contains a trace of performance metrics recorded at the end of each training epoch. This includes the chosen loss function and each configured metric, such as accuracy.
 - A learning curve is a plot of the loss/metric on the training dataset and the validation dataset. The plot can be created from the `History` object using the `Matplotlib` library.

Diagnosing Model Behavior

Plotting Model Learning Curves:

```
import matplotlib.pyplot as plt
history = model.fit(x_train, y_train, epochs=150, batch_size=32, verbose=0, validation_data=(x_test,y_test))

# Plot training & validation accuracy values
plt.plot(history.history['acc'])
plt.plot(history.history['val_acc'])
plt.title('Model accuracy')
plt.ylabel('Accuracy')
plt.xlabel('Epoch')
plt.legend(['Train', 'Test'], loc='upper left')
plt.show()

# Plot training & validation loss values
plt.plot(history.history['loss'])
plt.plot(history.history['val_loss'])
plt.title('Model loss')
plt.ylabel('Loss')
plt.xlabel('Epoch')
plt.legend(['Train', 'Test'], loc='upper left')
plt.show()
```

Deep Learning Model Life-Cycle

Analysing performance of ANN



Neural Network:

Diagnosing Model Behaviour

Diagnosing Model Behaviour

- ❑ The shape and dynamics of a learning curve can be used to diagnose the behavior of a Neural Network.
- ❑ There are 3 common conditions that can be observed:
 - Underfit
 - Overfit
 - Good Fit

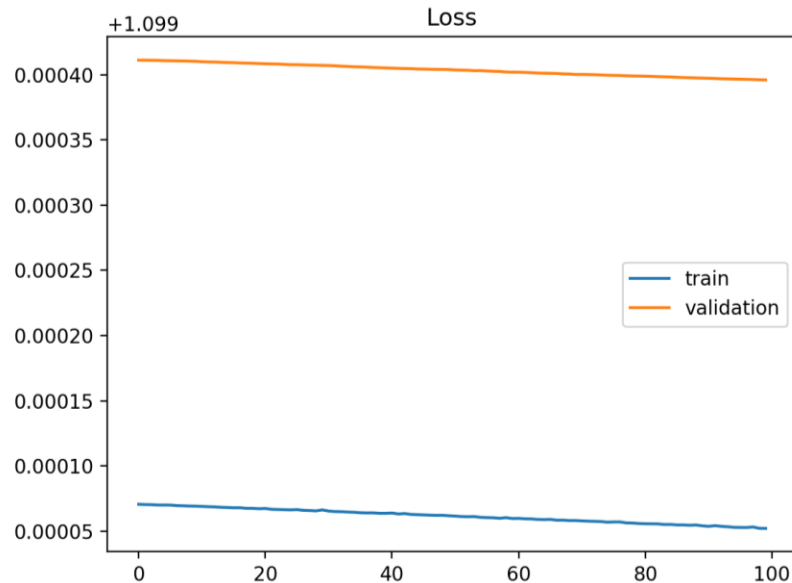
Diagnosing Model Behaviour

☐ Underfit Learning Curves

- ☐ Underfitting refers to a model that cannot learn the training dataset.
- ☐ An underfit model can be identified from the learning curve of the training loss only.
- ☐ It may show a flat line or noisy values of relatively high loss, indicating that the model was unable to learn the training dataset at all.
- ☐ Underfit is common when the model does not have a suitable capacity for the complexity of the dataset.

Diagnosing Model Behaviour

- ❑ **Underfit Learning Curves:** The training loss remains flat regardless of training.



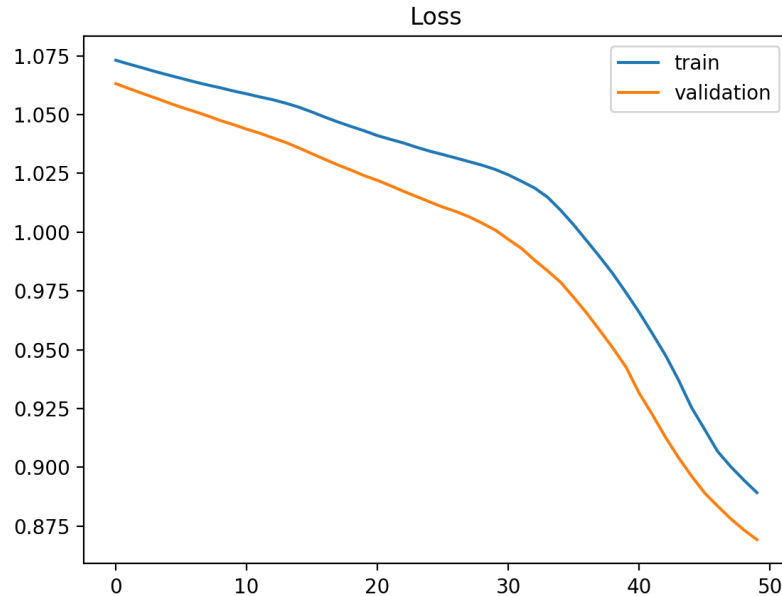
Diagnosing Model Behaviour

☐ Underfit Learning Curves

- ☐ An underfit model may also be identified by a training loss that is decreasing and continues to decrease at the end of the plot.
- ☐ This indicates that the model is capable of further learning and possible further improvements and that the training process was halted prematurely.

Diagnosing Model Behaviour

- Underfit Learning Curves: The training loss continues to decrease until the end of training.



Diagnosing Model Behaviour

☐ Overfit Learning Curves

- ☐ Overfitting refers to a model that has learned the training dataset too well, including the statistical noise or random fluctuations in the training dataset.
- ☐ The more specialized the model becomes to training data, the less well it is able to generalize to new data, resulting in an increase in generalization error.

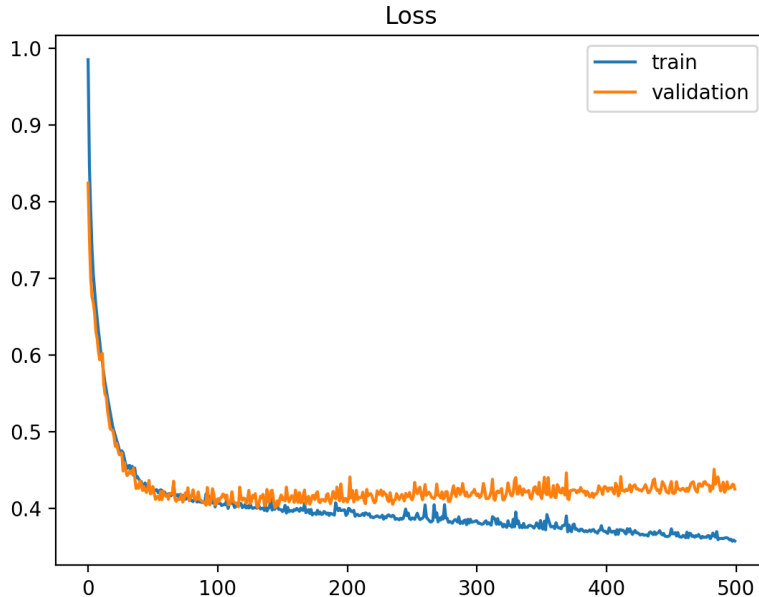
Diagnosing Model Behaviour

☐ Overfit Learning Curves

- ☐ This increase in generalization error can be measured by the performance of the model on the validation dataset.
- ☐ This often occurs if the model has more capacity than is required for the problem, and, in turn, too much flexibility. It can also occur if the model is trained for too long.

Diagnosing Model Behaviour

Overfit Learning Curves



A plot of learning curves shows overfitting if:

- ❑ The plot of training loss continues to decrease with experience.
- ❑ The plot of validation loss decreases to a point and begins increasing again.

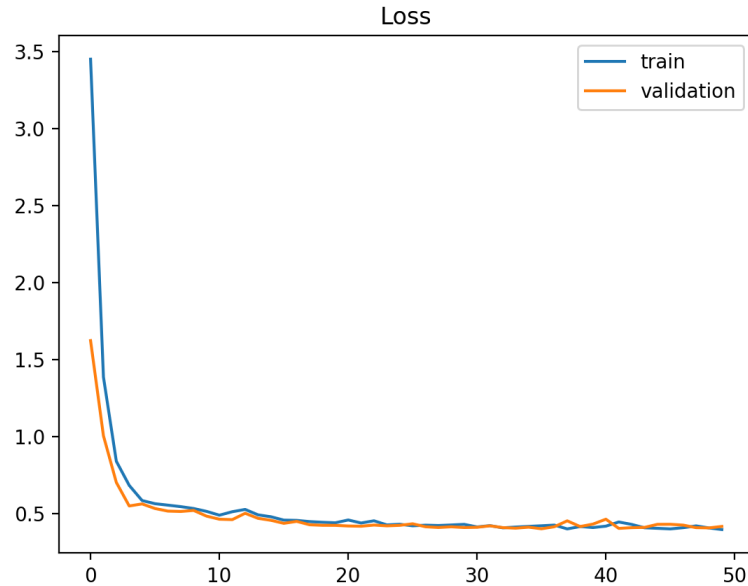
Diagnosing Model Behaviour

☐ Good Fit Learning Curves

- ☐ A good fit is the goal of the learning algorithm and exists between an overfit and underfit model.
- ☐ A good fit is identified by a training and validation loss that decreases to a point of stability with a minimal gap between the two final loss values.

Diagnosing Model Behaviour

Good Fit Learning Curves



Diagnosing Model Behaviour

☐ Unrepresentative Train Dataset

- ☐ Learning curves can also be used to diagnose properties of a dataset and whether it is relatively representative.
- ☐ An unrepresentative training dataset means that the training dataset does not provide sufficient information to learn the problem, with respect to the validation dataset used to evaluate it.

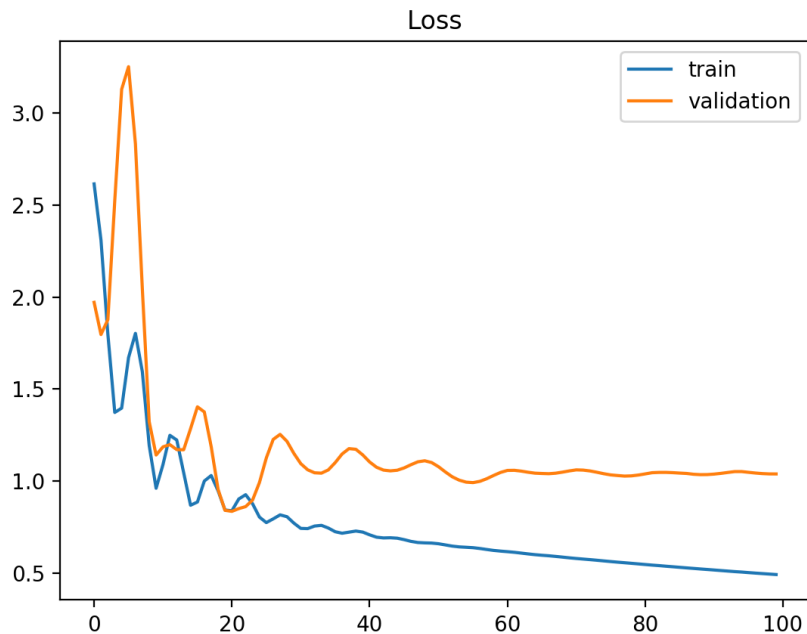
Diagnosing Model Behaviour

☐ Unrepresentative Train Dataset

- ☐ This may occur if the training dataset has too few examples as compared to the validation dataset.
- ☐ This situation can be identified by a learning curve for training loss that shows improvement and similarly a learning curve for validation loss that shows improvement, but a large gap remains between both curves.

Diagnosing Model Behaviour

□ Unrepresentative Train Dataset



Coming next:

Challenges with training ANNs